

To Search or Not to Search: Aligning the Decision Boundary of Deep Search Agents via Causal Intervention

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Abstract

Deep search agents, which autonomously iterate through multi-turn web-based reasoning, represent a promising paradigm for complex information-seeking tasks. However, current agents suffer from critical inefficiency: they conduct excessive searches as they cannot accurately judge when to stop searching and start answering. This stems from outcome-centric training that prioritize final results over the search process itself. We identify the root cause as misaligned **decision boundaries**, the threshold determining when accumulated information suffices to answer. This causes **over-search** (redundant searching despite sufficient knowledge) and **under-search** (premature termination yielding incorrect answers). To address these errors, we propose a comprehensive framework comprising two key components. First, we introduce **causal intervention-based diagnosis** that identifies boundary errors by comparing factual and counterfactual trajectories at each decision point. Second, we develop **Decision Boundary Alignment** for Deep Search agents (**DAS**), which constructs preference datasets from causal feedback and aligns policies via preference optimization. Experiments on public datasets demonstrate that decision boundary errors are pervasive across state-of-the-art agents. Our DAS method effectively calibrates these boundaries, mitigating both

over-search and under-search to achieve substantial gains in accuracy and efficiency. Our code and data are publicly available at: https://github.com/Applied-Machine-Learning-Lab/WWW2026_DAS.

CCS Concepts

• **Information systems** → **Information retrieval**.

Keywords

Deep Search Agent, Large Language Models

ACM Reference Format:

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1 Introduction

The integration of Large Language Models (LLMs) with autonomous agents has revolutionized intelligent information systems. Beyond text generation, agent-based paradigms are now reconstructing behavior spaces [37, 44, 56, 59], optimizing sequential decisions [7, 27, 61–63], enhancing ranking tasks [24], and advancing personalization [23]. Focusing on complex information seeking, Agentic RAG is emerging as a promising paradigm [10, 25]. Unlike traditional RAG systems, it enables sophisticated interactions with search tools [50], including adaptive search invocation, iterative query decomposition and refinement [3], systematic validation of retrieved content, and resolution of conflicts between parametric and web knowledge [10, 22, 60]. While such retrieval frameworks can also incorporate knowledge graphs [49, 52] and multimodal

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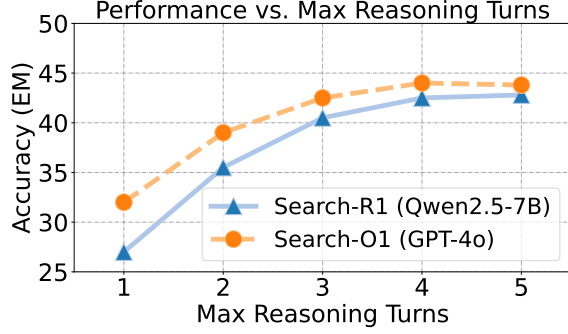


Figure 1: Trade-off between search depth and accuracy on HotpotQA, illustrating the diminishing gains.

information [4, 5] to enhance RAG performance [48], the critical advancement lies in the autonomous search capability. Recent reasoning LLMs have begun to internalize these search capabilities [8, 51], enabling the development of *deep search agents* [6, 13, 15, 16] that support complex reasoning and in-depth research workflows.

Despite their promise, current deep search agents lack efficiency, often conducting excessive search rounds to solve tasks. This inefficiency stems from outcome-centric training and evaluation paradigms that prioritize final results over intermediate decision quality [15, 16, 39, 45]. By treating the reasoning process as a black box, these approaches incentivize deep search agents to search repeatedly as long as the task is eventually solved, without considering the decision optimality of individual round: *whether to continue searching or to answer*. However, our empirical study reveals that search gains diminish as rounds increase, which is especially significant after 3 rounds as shown in Figure 1. While accuracy improves with more reasoning turns, the marginal benefit of each additional search decreases substantially. This finding underscores a critical challenge: determining the optimal decision boundary that identifies when to stop searching.

We formally define the *search decision boundary* (§2.1) as the critical threshold determining when parametric knowledge and accumulated retrieved information, becomes sufficient to answer the query. This boundary is pivotal for balancing the accuracy-efficiency trade-off: while additional search rounds may improve accuracy, they incur higher latency and computational costs. However, current deep search agents struggle to optimize this boundary, leading to two prevalent failure modes. **Over-search** occurs when agents conduct redundant searches despite possessing sufficient information, driven by over-reliance on external tools and neglecting internal knowledge [42]. Methods like WebResearcher [32], WebSailor [20], and DeepDive [28] achieve performance gains through extensive tool use but suffer from this “*cognitive offloading*” [36], failing to prevent unnecessary searches. Conversely, **under-search** occurs when agents prematurely terminate search and provide incorrect answers. Knowledge boundary-based approaches like FLARE [14] and Adaptive RAG [12] attempt to prioritize internal knowledge through confidence estimation and routing strategies. While they can successfully reduce the usage of search tools, they

tend to be over-confident on internal knowledge, leading to premature search termination. Neither paradigm effectively aligns the search decision boundary to optimize both accuracy and efficiency.

To address these decision boundary errors, we propose a comprehensive framework that systematically diagnoses and corrects suboptimal search decisions through two key components. First, we introduce **causal intervention-based diagnostic methodology** (§2.2) that identifies both over-search and under-search errors by comparing an agent’s factual trajectory against counterfactual alternatives. Specifically, when an agent searches, we simulate the outcome had it answered instead; when it answers incorrectly, we simulate the outcome of continued search. By evaluating these counterfactual outcomes, we can retrospectively determine whether each decision was causally optimal given the agent’s latent knowledge state. Second, building upon this diagnostic foundation, we develop the **Decision Boundary Alignment (DAS)** (§2.3). DAS constructs targeted preference pairs from the causal feedback (via pairing suboptimal factual trajectories with their superior counterfactual counterparts) and aligns agent’s policy via preference optimization. This alignment process directly teaches agent to internalize when its knowledge is sufficient to answer versus when further search is needed. Extensive experiments on three public datasets demonstrate that DAS can successfully calibrate the decision boundary, reducing both over-search and under-search errors while significantly improving accuracy and efficiency. In summary, our contributions are in threefold:

- **Problem Formulation and Analysis.** We formally define and characterize the two primary decision boundary errors, over-search and under-search, and demonstrate through extensive experiments that they constitute a fundamental bottleneck in current deep search agents.
- **Causal Diagnostic Methodology.** We propose a novel diagnostic approach using *causal intervention* to rigorously identify both error types by comparing factual and counterfactual trajectories at each decision point.
- **Decision Boundary Alignment (DAS).** We introduce DAS, which leverages causal feedback to construct preference datasets and optimize searching policies, achieving substantial gains in both accuracy and efficiency.

2 Decision Boundary Detection and Alignment

This section presents our approach to align the agent’s decision-making with optimal decision preference. We formalize the decision boundary and define two key **decision errors: over-search and under-search** (§2.1). To mitigate these errors, we then introduce our method for decision boundary detection (§2.2) and our Decision Boundary Alignment for Search (DAS) algorithm (§2.3). Figure 2 illustrates the overall framework.

2.1 Formal Definition of the Decision Boundary

To rigorously assess the decision-making capabilities of a deep search agent, we move beyond holistic trajectory evaluations and adopt a causal lens focused on step-wise optimality. We model the decision process at step t as an interaction between the agent’s chosen action $A_t \in \{\text{Search}, \text{Answer}\}$ and the unobservable ground

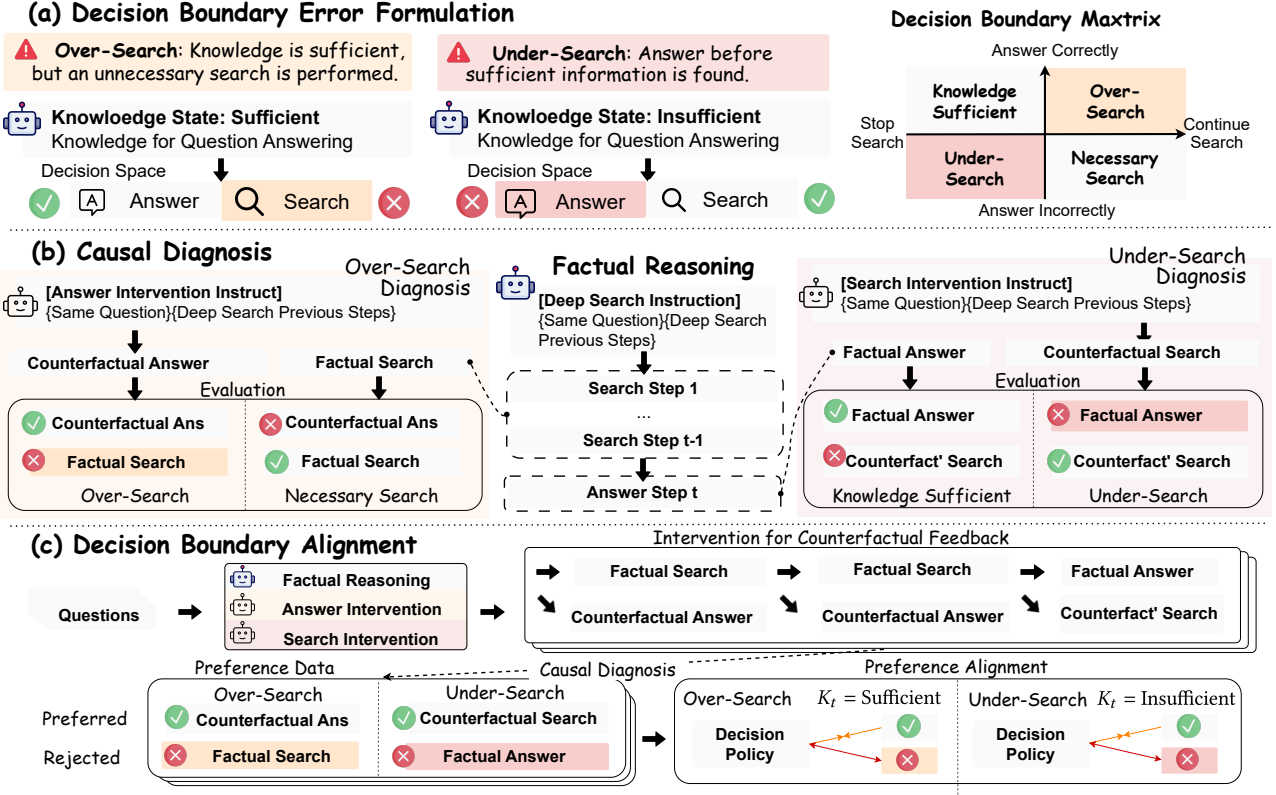


Figure 2: The overall framework of DAS. (a) Formal definition of over-search and under-search as the two primary failure modes at the agent’s decision boundary. (b) Causal Intervention for decision errors diagnosis within an agent’s trajectory through the use of counterfactual instruction. (c) Decision Boundary Alignment for Search Agent based on a preference dataset from identified errors.

truth of its knowledge state $K_t \in \{\text{Sufficient}, \text{Insufficient}\}$. As conceptually visualized in Figure 2(a), the decision boundary is not merely a theoretical line but the precise threshold where the agent must distinguish between the necessity of further exploration and the sufficiency of current information. We ground these quadrants in a formal utility framework to mathematically define the conditions that render specific actions suboptimal.

Utility and Error Formulation. The agent’s objective is to maximize the expected utility $U(A_t, K_t)$, which is governed by a set of rewards and costs. We posit that a correct answer yields a high positive reward R_{correct} , whereas providing an incorrect answer incurs a significant penalty $R_{\text{incorrect}}$. Conversely, the act of searching is nuanced: it incurs a computational and operational cost C_{search} , but offers a positive information value R_{info} when the current knowledge is lacking. This formulation allows us to categorize decision failures not just as wrong outcomes, but as specific deviations from optimal utility:

- **Over-Search (Efficiency Failure):** An over-search error is strictly defined by the state where the agent chooses to Search despite K_t already being Sufficient. In this scenario, the search action is mathematically inferior because the information gain is zero, yet

the cost C_{search} is still incurred. The alternative action, Answer, would have immediately yielded the maximum reward R_{correct} without further expense. Thus, over-search represents a failure of **efficiency**, where the agent sacrifices immediate operational success for unnecessary computational effort **that provides no marginal benefit to the answer’s quality**.

- **Under-Search (Accuracy Failure):** Conversely, an under-search error occurs when the agent chooses to Answer while K_t remains Insufficient. This decision is catastrophic, as the premature termination of the search process triggers the penalty $R_{\text{incorrect}}$. Here, the alternative Search action is the dominant strategy because the value of acquiring missing information (R_{info}) logically outweighs the search cost (C_{search}). By failing to recognize the need for further exploration, the agent compromises the factual validity of its response. This represents a fundamental **failure of accuracy** driven by a flawed assumption of sufficiency, ignoring that the prospective payoff from searching is rationally superior.

This causal framework transforms the decision-making process into an optimization problem. The agent’s goal is to align its policy $\pi(S_t)$ with the latent state K_t , ensuring it selects the action that mathematically dominates the alternative in each scenario.

2.2 Causal Diagnosis of Decision Boundary Errors

Localizing decision boundary errors is challenging since unselected actions cannot be observed. To diagnose the decision errors defined by the unobservable latent state K_t , we employ causal intervention, a principled method from causality theory for estimating the counterfactual outcomes of alternative actions [30]. The diagnostic process is retrospective. Each decision point t is revisited after an agent's trajectory is complete to apply an intervention, which is formalized by the $do()$ -operator as $do(A_t := a')$. The resulting counterfactual outcome is then compared against the factual one to infer the value of K_t and identify the error. The causal diagnostic process is now detailed for each of the two error types: over-search and under-search.

Diagnosing Over-search. An over-search error is diagnosed at a state S_t when the agent's factual action is $a_t = \text{Search}$. The diagnosis hinges on determining if the latent state was actually $K_t = \text{Sufficient}$. To resolve this, the causal intervention $do(A_t := \text{Answer})$ is applied to simulate the alternative scenario. A key insight is that a definitive diagnosis simplifies to a practical test of the counterfactual answer's quality, sidestepping complex utility calculations. A correct counterfactual answer implies that the agent's knowledge state was indeed sufficient, whereas an incorrect answer implies it was not.

The $do(A_t := \text{Answer})$ -operator is implemented via the answer intervention instruction. This instruction is designed to compel the agent to synthesize a final answer using only its current knowledge state S_t . The detailed prompt design is provided in Appendix B.

Diagnosing Under-search. In the case of an under-search error, the diagnosis is revealed by the factual outcome. When the agent's action is $A_t = \text{Answer}$ and the resulting answer is inaccurate, the latent state is immediately inferred to be $K_t = \text{Insufficient}$. No intervention is needed to identify the error itself.

Intervention is nonetheless critical for the subsequent task of remediation, which is to generate a corrected trajectory for the alignment dataset. This task addresses the counterfactual question of what the superior outcome would have been had the agent searched instead. The intervention $do(A_t := \text{Search})$ is therefore applied to generate this outcome. It is implemented via the search intervention instruction, which guides the agent to execute its higher-utility action. The resulting corrected, counterfactual trajectory then serves as the positive example for aligning the agent's decision policy. The detailed prompt design is provided in Appendix B.

2.3 Decision Boundary Alignment from Causal Feedback

While causal diagnosis reveals where the agent's decisions deviate from optimality, effective improvement requires aligning its policy with these causal signals to refine the decision boundary. The causal interventions performed during diagnosis do more than identify errors; they generate counterfactual trajectories that provide a powerful signal for policy alignment. By comparing the agent's suboptimal factual path with the superior counterfactual path revealed through intervention, we obtain a clear causal preference.

This section details the construction of a targeted preference dataset from these signals to align the agent's decision policy, π .

The dataset is composed of preference pairs, (preferred sequence y_c , rejected sequence y_r), derived directly from the diagnosed decision boundary failures.

Preferences from Over-search Errors. When an over-search error at state S_t is diagnosed, the intervention $do(A_t := \text{Answer})$ reveals a more efficient path to a correct answer. For this case, a preference pair is formed to reward efficiency:

- The **preferred sequence** (y_c) is the shorter, counterfactual trajectory generated by the intervention, which directly answers from state S_t .
- The **rejected sequence** (y_r) is the agent's original, factual trajectory, which includes the causally redundant Search step.

Preference from Under-search Errors. When an under-search error is diagnosed, the intervention $do(A_t := \text{Search})$ is applied to generate a corrective, higher-utility trajectory. This forms a preference pair that rewards thoroughness:

- The **preferred sequence** (y_c) is the superior, counterfactual trajectory resulting from the intervention, which includes the necessary search step.
- The **rejected sequence** (y_r) is the agent's original, factual trajectory that terminated prematurely with a low-quality answer.

The resulting preference dataset, $\mathcal{D} = \{(x, y_c, y_r)\}$, where x represents the state history leading to the decision, is used to fine-tune the policy π_θ by minimizing the Direct Preference Optimization [35] loss:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_c, y_r) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_c|x) \pi_{\text{ref}}(y_r|x)}{\pi_\theta(y_r|x) \pi_{\text{ref}}(y_c|x)} \right) \right] \quad (1)$$

where π_{ref} is the original agent policy, β is a temperature parameter, and σ is the logistic function. This training process incentivizes the policy to adopt the preferred, causally optimal decision paths while penalizing the suboptimal ones, thereby directly refining its judgment at the decision boundary.

3 Experiments

This section presents a comprehensive experimental evaluation conducted on three question-answering (QA) datasets. Our experiments are designed to answer the following research questions (RQs) systematically:

- **RQ1:** To what extent do decision boundary errors exist in current deep search agents?
- **RQ2:** Which question characteristics are more likely to cause decision boundary errors?
- **RQ3:** What is the impact of our proposed Decision Boundary Alignment on agent accuracy, efficiency, and decision quality?
- **RQ4:** What is the relationship and discrepancy between an agent's knowledge boundary and its decision boundary?
- **RQ5:** How does the number of reasoning steps affect the prevalence of decision boundary errors?

In the following subsections, we first detail our experimental setup, after which dedicated analyses are provided to address each of these research questions.

3.1 Experimental Settings

3.1.1 Dataset. Our experiments are conducted on three benchmark datasets selected to assess performance across varying levels of reasoning complexity. We use **Natural Questions (NQ)** [18] to evaluate single-hop retrieval performance. For multi-hop reasoning, we employ **HotpotQA** [53] and **2WikiMultiHopQA** [9], both of which require aggregating information from multiple documents. We adopt the standard settings for all datasets, utilizing the official training and development splits.

3.1.2 Baselines. Our evaluation is designed to first validate the existence of decision boundary errors and then to assess our proposed alignment method. To this end, we employ two distinct and representative workflows. Search-R1 embodies the approach of using end-to-end reinforcement learning (RL) to directly optimize a search agent. This workflow serves as our primary testbed, where we compare different models from the Qwen series [40] (Qwen2.5-7B/14B-base vs. -RL) and, crucially, evaluate our Decision Boundary Alignment. In contrast, Search-O1 represents the paradigm of integrating search tools into powerful general-purpose models. This allows us to analyze the behavior of leading closed-source models, including OpenAI’s GPT-4o [11], GPT-4o-mini, Google’s Gemini-2.5-Flash [2], and DeepSeek V3 [26], in a search-augmented setting.

3.1.3 Metrics. Our evaluation protocol covers three critical aspects of agent performance. First, to assess the fundamental accuracy, we utilize the **Exact Match (EM)** metric. Second, considering the practical efficiency of agents, we evaluate efficiency using **Total Inference Time** and the **Average Search Queries (ASQ)**, aiming for high accuracy with minimal retrieval costs. Finally, to provide a granular analysis of the agent’s decision boundary errors, we calculate the **Over-Search Rate (OSR)** and **Under-Search Rate (USR)**.

3.1.4 Implementation Details. Our retrieval pipeline is built upon a corpus from the Wikipedia dump (Wikidump-20250901 [46]), which is segmented into sentence-level chunks. For retrieval, we employ the E5 embedding model [43] to fetch the top-3 most relevant chunks per query. Our baseline Search-R1 workflow follows the implementation of the open-source FlashRAG framework. To perform our Decision Boundary Alignment, we constructed a preference dataset of 20,000 examples by sampling 10,000 instances from each of the NQ and HotpotQA training sets. This preference data was generated using the RL-tuned Qwen2.5-7B-RL and Qwen2.5-14B-RL models, which were then further fine-tuned on this same dataset. We perform the alignment training via DPO for 3 epochs. Key hyperparameters include a DPO β of 0.3, and a LoRA configuration [33] with rank r of 64 and α of 128 for parameter-efficient tuning.

3.2 Decision Boundary Analysis (RQ1)

We conducted extensive experiments to investigate the decision boundary problem across various deep search agents and dataset settings to answer RQ1. Specifically, We evaluated two primary categories of state-of-the-art agents. The first is Search-R1, which employs outcome-based reinforcement learning to optimize search capabilities. For this category, we compare the Qwen2.5-7B and Qwen2.5-14B models before and after RL fine-tuning. The second category, Search-O1, includes powerful, closed-source models such

as the GPT series, which serve as strong baseline agents. The comprehensive results are presented in Table 1, from which we derive the following key conclusions.

The Decision Boundary Problem is Pervasive. The results in Table 1 unequivocally demonstrate that flawed decision-making is a pervasive challenge. All tested agents, regardless of architecture, scale, or training paradigm, exhibit significant levels of both OSR and USR. For instance, even a powerful model like GPT-4o shows a USR of 0.670 on NQ, indicating a frequent failure to initiate a search when necessary. This establishes the decision boundary problem not as a niche issue but as a fundamental bottleneck for current search agents. These errors directly degrade performance: a high USR limits an agent’s potential accuracy by failing to retrieve critical information, while a high OSR harms efficiency by wasting computational resources on unnecessary queries.

Outcome-based RL Reveals a Critical Trade-off. We examined whether outcome-based RL mitigates decision boundary errors and found that it introduces a critical trade-off. On the NQ dataset, RL dramatically boosts the accuracy of Qwen2.5-7B (EM from 0.190 to 0.410), primarily by teaching the agent to be more thorough. This is reflected in a substantial reduction in USR (from 0.672 to 0.505) as the agent learns to search more often. However, this aggressive search policy comes at a steep efficiency cost: the OSR increases (from 0.196 to 0.263), and the average search queries (ASQ) rise by over 60%. This finding reveals that relying on accuracy-centric rewards can correct the critical issue of under-search, but at the expense of pushing the decision boundary too far. By incentivizing correctness without penalizing cost, the model alleviates under-search only to exacerbate over-search.

Universality Across Scales and Workflows. Our empirical analysis indicates that the challenge of optimal decision boundary placement is pervasive, persisting across varying model scales and distinct workflows. While scaling up model parameters (e.g., from Qwen2.5-7B to 14B) improves general capabilities, Table 1 shows that it does not resolve the conflict between efficiency and accuracy: larger models still exhibit significant rates of both over-search and under-search. Furthermore, this trade-off transcends specific pipelines. Whether employing the training-centric Search-R1 workflow or the inference-centric Search-O1 workflow (including GPT-4o, Gemini-2.5-Flash, and DeepSeek-V3), no agent achieves a balance where both errors are negligible. This universality suggests that the bottleneck is not merely model capability, but the fundamental difficulty of calibrating the optimal decision boundary.

Influence of Task Complexity. Finally, comparing performance on NQ versus the multi-hop HotpotQA dataset highlights how task complexity exacerbates decision-making challenges. As expected, all agents increase their search frequency (higher ASQ) on the more complex HotpotQA task, showing a basic ability to adapt. However, this adaptation is imperfect. The persistently high OSR and USR values on HotpotQA indicate that as reasoning chains grow longer, an agent’s ability to precisely determine the necessity of a search at each intermediate step diminishes significantly.

Table 1: Performance comparison of eight models across different deep search agents on the NQ, HotpotQA, and 2WikiMultiHopQA datasets. ASQ denotes Average Search Queries.

Workflow	Model	NQ					HotpotQA					2WikiMultiHopQA				
		EM	Time	ASQ	OSR	USR	EM	Time	ASQ	OSR	USR	EM	Time	ASQ	OSR	USR
Search-R1	Qwen2.5-7b-base	0.174	269.77	1.142	0.1265	0.7651	0.197	316.05	1.817	0.1712	0.7167	0.253	352.44	2.342	0.2721	0.5793
	Qwen2.5-7b-RL	0.381	464.02	2.014	0.3280	0.5210	0.407	638.12	2.640	0.3020	0.5430	0.457	644.81	3.200	0.3920	0.4090
	Qwen2.5-14b-base	0.260	727.55	1.714	0.1930	0.6590	0.292	834.74	2.168	0.2202	0.6206	0.345	816.80	2.553	0.2358	0.5435
	Qwen2.5-14b-RL	0.440	331.19	1.016	0.3233	0.5015	0.421	540.58	1.457	0.2928	0.5216	0.569	768.11	2.168	0.3195	0.3630
Search-O1	GPT-4o	0.270	40.86	1.226	0.1473	0.6748	0.440	63.31	1.785	0.2592	0.4311	0.520	87.86	2.653	0.1581	0.4729
	GPT-4o-mini	0.170	325.65	1.230	0.1238	0.8314	0.360	134.97	1.772	0.1859	0.6407	0.290	130.83	2.308	0.1224	0.6805
	Gemini-2.5-Flash	0.350	198.44	0.591	0.1600	0.7198	0.390	281.86	0.404	0.1167	0.5415	0.390	210.69	0.390	0.2500	0.4952
	deepseek-V3	0.250	126.51	1.141	0.2906	0.6289	0.470	195.28	1.383	0.4374	0.4122	0.460	180.48	1.595	0.0218	0.5263

Table 2: Performance analysis across different data characteristics. We analyze the agent’s decision boundary errors based on question difficulty, category, and the required number of supporting facts.

Factor	Type	EM	F1	ACC	ASQ	OSR	USR
Difficulty Level	Easy	0.500	0.623	0.523	2.318	0.402	0.402
	Medium	0.506	0.570	0.529	2.674	0.394	0.423
	Hard	0.453	0.552	0.480	2.598	0.402	0.469
Question Category	Comparison	0.717	0.806	0.754	2.594	0.738	0.193
	Bridge	0.444	0.526	0.465	2.582	0.319	0.481
# Supporting Facts	2	0.520	0.604	0.546	2.526	0.423	0.399
	3	0.464	0.541	0.469	2.680	0.356	0.464
	4	0.360	0.445	0.413	2.867	0.293	0.560
	≥5	0.300	0.417	0.300	2.800	0.100	0.700

3.3 Impact of Data Characteristics on Decision Errors (RQ2)

We conduct extensive experiments to investigate the decision boundary errors on different type of datasets to answer RQ2. Specifically, We perform a fine-grained analysis of the agent’s performance on subsets of the HotpotQA dataset, partitioned along three distinct dimensions of complexity. This allows us to create controlled testbeds to isolate specific failure modes.

- **Difficulty Level:** This dimension categorizes questions as Easy, Medium, or Hard, based on human annotations established during the dataset’s creation. The level reflects the human-perceived difficulty of the search task and generally correlates with the number of search actions required.
- **Question Category:** This dimension distinguishes between two reasoning types. Comparison questions require retrieving multiple, independent pieces of information. In contrast, Bridge questions necessitate a sequential reasoning path where the queries are dependent on one another.
- **# Supporting Facts:** This metric refers to the number of distinct text paragraphs required to derive the final answer. It directly correlates with the searching burden and the minimum number of retrieval actions needed to complete the task.

The experimental results are summarized in Table 2, and our key observations are detailed below.

As the general difficulty increases from Easy to Hard, the agent’s failure mode shifts. While accuracy metrics (EM, F1, ACC) decline slightly, the most significant change is the steady increase in the USR from 0.402 to 0.469. The OSR and the average search queries (ASQ) remain relatively stable. This suggests that as questions become implicitly harder, the agent’s primary weakness is its failure to recognize the need for more rigorous information gathering. It is prone to halting the search process prematurely and attempting to answer with insufficient evidence.

Impact of Question Category. Comparison and Bridge questions exhibit two starkly different decision-making pathologies. Comparison questions, which involve parallel information retrieval, yield high accuracy (0.754 ACC) but suffer from an extremely high OSR of 0.738. This indicates that while the agent correctly identifies the need for multiple pieces of information, its decision process is highly inefficient, leading to redundant searches. Conversely, Bridge questions, which require a dependent reasoning path, show an inverse error pattern. The OSR is low (0.319), but the USR is alarmingly high at 0.481. This reveals a critical failure in multi-step reasoning: the agent struggles to use the result of one search to inform the necessity of the next. Its primary failure is not inefficient searching, but rather *not knowing it needs to continue searching at all*.

Impact of the Number of Supporting Facts. This dimension directly measures the information-gathering burden. As the required number of supporting facts increases from 2 to 5 or more, a clear pattern of insufficient persistence emerges. Although the agent slightly increases its search attempts (ASQ rises from 2.526 to 2.800), this is inadequate for the task’s escalating complexity. The result is a dramatic trade-off: OSR plummets from 0.423 to just 0.100, while the USR skyrockets from 0.399 to a critical 0.700. The low OSR for questions needing ≥5 facts is not a sign of efficiency; it is a symptom of the agent *giving up too early*. It executes initial searches but lacks the decision-making logic to persist until all necessary information is collected, highlighting a decision boundary biased towards premature termination on information-intensive questions.

3.4 Decision Boundary Alignment (RQ3)

This section evaluates the efficacy of our proposed decision boundary alignment method to answer RQ3. Specifically, we apply our

Table 3: Main results and ablation studies for our decision boundary alignment method applied to Search-R1. For each metric, the best-performing result is in bold and the second-best is underlined.

Model	NQ					HotpotQA					2WikiMultiHopQA				
	EM	Time	ASQ	OSR	USR	EM	Time	ASQ	OSR	USR	EM	Time	ASQ	OSR	USR
Qwen2.5-7B-RL	0.381	464.02	2.014	0.3280	0.5210	0.407	638.12	2.640	0.3020	0.5430	0.457	644.81	3.200	0.3920	0.4090
+ DAS	0.394	<u>432.06</u>	<u>1.896</u>	<u>0.3080</u>	<u>0.5180</u>	0.414	593.45	<u>2.568</u>	<u>0.2790</u>	<u>0.5370</u>	0.478	<u>632.71</u>	<u>3.113</u>	<u>0.3746</u>	<u>0.4030</u>
w/o under-search data	0.392	415.52	1.758	0.2883	0.5356	<u>0.412</u>	578.11	2.411	0.2595	0.5591	<u>0.476</u>	615.90	2.945	0.3528	0.4242
w/o over-search data	0.387	533.62	2.215	0.3580	0.4410	0.409	733.84	2.855	0.3320	0.4630	0.461	741.53	3.410	0.4220	0.3290
Qwen2.5-14B-RL	0.440	331.19	1.016	0.3233	0.5015	0.421	540.58	1.457	0.2928	0.5216	0.569	768.11	2.168	0.3195	0.3630
+ DAS	0.454	<u>308.02</u>	<u>0.955</u>	<u>0.3039</u>	<u>0.4965</u>	0.428	502.74	<u>1.413</u>	<u>0.2723</u>	<u>0.5164</u>	0.580	<u>714.34</u>	<u>2.103</u>	<u>0.2971</u>	<u>0.3594</u>
w/o under-search data	0.452	295.18	0.861	0.2854	0.5147	<u>0.426</u>	489.25	1.302	0.2541	0.5388	<u>0.578</u>	698.55	1.986	0.2798	0.3776
w/o over-search data	0.446	380.87	1.251	0.3533	0.4781	0.423	621.67	1.682	0.3228	0.4952	0.572	883.33	2.455	0.3495	0.2830

alignment technique to two deep search agents: Qwen2.5-7B-RL and Qwen2.5-14B-RL, both trained using Search-R1. To dissect the contribution of each component, we conduct an ablation study with the following variants:

- **DAS:** Our proposed Decision Boundary Alignment method, which aligns the model using preference data that targets both under-search and over-search errors.
- **w/o under-search data:** A variant that omits under-search data, relying solely on over-search preference data for alignment.
- **w/o over-search data:** A variant that omits over-search data, relying solely on under-search preference data for alignment.

The experimental results are summarized in Table 3 and the conclusion are listed as follows.

Main Alignment Results. As demonstrated in Table 3, our proposed DAS method yields substantial and consistent improvements. This performance gain is robust across both the Qwen2.5-7B-RL and Qwen2.5-14B-RL models and holds for all three QA datasets. Specifically, DAS achieves the highest accuracy, as measured by exact match (EM), in nearly all configurations. Furthermore, the method enhances operational efficiency by significantly reducing both inference time and the average number of search queries (ASQ) compared to the baseline RL models. Crucially, DAS effectively mitigates decision boundary errors by simultaneously lowering the over-searching rate (OSR) and the under-searching rate (USR), resulting in a more calibrated and reliable search agent.

Ablation Study. Our ablation study dissects the contribution of each data component, revealing a critical trade-off. Removing the under-search preference data (*w/o under-search data*) compels the agent to adopt a highly aggressive and efficient policy, achieving the best scores on all efficiency-related metrics (time, ASQ, and OSR). Conversely, omitting the over-search data (*w/o over-search data*) induces an overly cautious agent. This variant exhibits a stronger inclination to search, minimizing the USR but at the expense of overall efficiency. These two signals function antagonistically; emphasizing one degrades performance in the other’s domain. Therefore, only the complete DAS framework, which concurrently balances both error types, achieves a superior synthesis of top-tier accuracy and well-rounded efficiency.

Table 4: Correlation between search decisions and knowledge boundary. Acc measures the agent’s accuracy when forced to answer without searching (Probe 1). Confidence measures the frequency (% ‘Yes’) the agent claims to know the answer (Probe 2).

Decision Category	Proportion	ACC	Confidence
Over-search	19.02%	0.2356	0.5750
Effective search	80.98%	0.1423	0.4859

3.5 Analysis of the Knowledge-Decision Gap (RQ4)

The relationship between an agent’s knowledge boundary and its decision boundary remains an open question. To investigate the gap between them(RQ4), we categorize agent searches into two types: Over-search (unnecessary) and Effective search (beneficial). We then probe the agent’s knowledge using two metrics: *ACC*, its accuracy when intervened to answer without searching, and *Confidence*, its self-assessed ability to answer (‘Yes/No’). The full prompts used in our study are available in Appendix B.

Table 4 presents our findings. The results reveal a nuanced picture. We observed that Over-search, which constitutes 19.02% of all actions, paradoxically shows higher agent ACC (0.2356 vs. 0.1423) and Confidence (0.5750 vs. 0.4859) compared to Effective search. This leads to a two-part conclusion. First, a correlation exists, as the agent’s knowledge state clearly influences its search decisions. However, the low absolute ACC highlights a severe miscalibration: the agent still fails to answer correctly in over 76% of these “unnecessary” search cases. We conclude that the Knowledge-Decision Gap stems from the agent’s poor self-awareness, which remains a key bottleneck for current architectures.

3.6 Step-wise Analysis of Decision Errors (RQ5)

To dissect the dynamics of the **decision boundary problem** during an agent’s autonomous reasoning process, we performed a step-wise analysis. This approach allows us to observe how the two core facets of this problem, **OSR** and **USR**, are distributed at each step. By comparing single-hop and multi-hop tasks, we aim to reveal how the dominant failure mode of the decision boundary evolves as

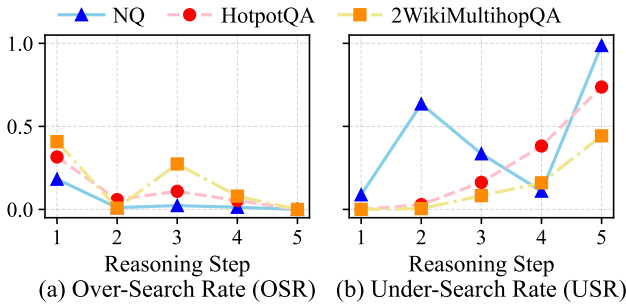


Figure 3: Distribution of decision boundary errors (OSR and USR) across reasoning steps on NQ, HotpotQA, and 2WikiMultihopQA (2Wiki) datasets.

the reasoning chain lengthens and task complexity increases. The results, presented in Figure 3, reveal distinct and evolving patterns.

Over-search is Concentrated in the Initial Step. A universal phenomenon across all datasets is that the OSR is highest at the first reasoning step. This indicates a general failure mode where agents reflexively initiate a search, even for questions that could potentially be answered directly from their parametric knowledge. While the OSR trends downwards significantly in subsequent steps, complex multi-hop tasks like 2WikiMultihopQA can exhibit a mid-process resurgence of over-search (0.274 at Step 3). This suggests that even when an agent correctly decides to continue its search, its mid-process decisions can still be inefficient and confused.

Under-search Escalates with Reasoning Steps. In contrast to OSR, the Under-search Rate (USR) systematically increases with the number of reasoning steps across all datasets. This indicates that agents are progressively more likely to answer incorrectly as task complexity increases, consistently misjudging information sufficiency for more difficult problems.

Divergent Error Trajectories for Single-hop vs. Multi-hop Tasks. The error patterns diverge significantly between task types. The single-hop NQ dataset shows an abrupt transition, where the error mode flips from a high OSR at step one to a critically high USR immediately after. In contrast, multi-hop datasets exhibit a gradual deterioration, with the USR climbing steadily at each subsequent step, indicating a progressive failure to assess information completeness.

4 Related Work

4.1 Knowledge Boundaries in Agents

Research on the *Knowledge Boundary* of LLMs aims to delineate the scope of their reliable parametric knowledge to mitigate hallucinations [21, 29, 54]. Boundary detection is often approached by quantifying model uncertainty via internal metrics like token entropy [17, 34] or by probing for model self-knowledge [55]. To mitigate this boundary problem, RAG method augments the context with documents retrieved from external corpora, extending the knowledge available during the inference stage [19]. However, the conventional RAG paradigm relies on a static retrieval mechanism,

making it inefficient for complex tasks that demand iterative reasoning rather than simple one-off information fetching [1]. This limitation becomes a critical bottleneck for deep search agents, which require autonomous, multi-step interaction with web search engines [15, 22]. For such agents, the problem evolves from knowledge retrieval to a sequential decision-making challenge. The central research question thus shifts from the static *Knowledge Boundary* (i.e., whether a model *can* answer) to a dynamic *Decision Boundary* (i.e., whether an agent *should* continue to search). Optimizing this decision policy is the core challenge we address.

4.2 Deep Search Agents

Deep search agents are autonomous systems for addressing complex questions through multi-step information exploration and synthesis [10, 25, 47]. Current research aims to enhance their dynamic decision-making and query generation [39, 57]. For instance, Search-o1 [22] employs structured iterative workflows to guide the agent’s reasoning process. Other works like Search-R1 [15], R1-Searcher [38], and ReasonRAG [58] train the end-to-end reasoning policy using final answer accuracy as a reward signal. InForage [31] and OTC [41] introduce heuristics like penalizing numerous search calls to curb inefficiency. However, prior work’s outcome-centric focus treats the reasoning process as a black box, overlooking the optimality of each intermediate decision. Relying on sparse final rewards, these approaches fail to distinguish essential exploration from redundant querying. This lack of granular supervision risks aimless information accumulation, leaving the critical problem of search termination unaddressed. Our work confronts this by formalizing it as the decision boundary challenge, learning an explicit policy to determine when information is sufficient.

5 Conclusion

In this paper, we formally define and address the *Decision Boundary* problem in deep search agents, a critical challenge causing failures of *over-search* (inefficiency) and *under-search* (inaccuracy). We introduce a novel diagnostic method rooted in causal intervention. This approach retrospectively infers an agent’s latent knowledge state by comparing its factual trajectory against a superior counterfactual one. Building on this causal diagnosis, we propose the **Decision Boundary Alignment (DAS)** method, which calibrates an agent’s judgment by constructing a preference dataset from these factual and counterfactual pairs. Our experiments provide the first quantitative evidence that these decision errors are a pervasive bottleneck in advanced agents. Crucially, the results demonstrate that DAS effectively mitigates both error types, achieving significant simultaneous gains in final-answer accuracy and overall efficiency. This research offers a foundational solution for creating more robust and economically viable autonomous agents for the web.

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References

- [1] Akari Asai, Zequi Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. In *International Conference on Learning Representations*.
- [2] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261* (2025).
- [3] Xingjian Diao, Chunhui Zhang, Weiyi Wu, Zhongyu Ouyang, Peijun Qing, Ming Cheng, Soroush Vosoughi, and Jiang Gui. 2025. Temporal working memory: Query-guided segment refinement for enhanced multimodal understanding. *arXiv preprint arXiv:2502.06020* (2025).
- [4] Kuicai Dong, Yujing Chang, Derrick Goh Xin Deik, Dexun Li, Ruiming Tang, and Yong Liu. 2025. MMDocLR: Benchmarking Multimodal Retrieval for Long Documents. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 30971–31005.
- [5] Kuicai Dong, Yujing Chang, Shijie Huang, Yasheng Wang, Ruiming Tang, and Yong Liu. 2025. Benchmarking Retrieval-Augmented Multimodal Generation for Document Question Answering.
- [6] Kuicai Dong, Shurui Huang, Fangda Ye, Wei Han, Zhi Zhang, Dexun Li, Wenjun Li, Qu Yang, Gang Wang, Yichao Wang, Chen Zhang, and Yong Liu. 2025. Doc-Researcher: A Unified System for Multimodal Document Parsing and Deep Research.
- [7] Zichuan Fu, Xiangyang Li, Chuhan Wu, Yichao Wang, Kuicai Dong, Xiangyu Zhao, Mengchen Zhao, Huifeng Guo, and Ruiming Tang. 2023. A unified framework for multi-domain ctr prediction via large language models. *ACM Transactions on Information Systems* (2023).
- [8] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirog Ma, Xiao Bi, et al. 2025. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* 645, 8081 (2025), 633–638.
- [9] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing A Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps. In *Proceedings of the 28th International Conference on Computational Linguistics*. 6609–6625.
- [10] Yuxuan Huang, Yihang Chen, Haozheng Zhang, Kang Li, Huichi Zhou, Meng Fang, Linyi Yang, Xiaoguang Li, Lifeng Shang, Songcen Xu, et al. 2025. Deep research agents: A systematic examination and roadmap. *arXiv preprint arXiv:2506.18096* (2025).
- [11] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276* (2024).
- [12] Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C Park. 2024. Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 7029–7043.
- [13] Pengcheng Jiang, Jiacheng Lin, Lang Cao, Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, and Jiawei Han. 2025. Deepretrieval: Hacking real search engines and retrievers with large language models via reinforcement learning. *arXiv preprint arXiv:2503.00223* (2025).
- [14] Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 7969–7992.
- [15] Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Serkan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516* (2025).
- [16] Can Jin, Yang Zhou, Qixin Zhang, Hongwu Peng, Di Zhang, Marco Pavone, Ligong Han, Zhang-Wei Hong, Tong Che, and Dimitris N Metaxas. 2025. Your reward function for rl is your best prm for search: Unifying rl and search-based tts. *arXiv preprint arXiv:2508.14313* (2025).
- [17] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221* (2022).
- [18] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics* 7 (2019), 453–466.
- [19] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.
- [20] Kuan Li, Zhongwang Zhang, Huifeng Yin, Liwen Zhang, Litu Ou, Jialong Wu, Wenbiao Yin, Baixuan Li, Zhengwei Tao, Xinyu Wang, et al. 2025. WebSailor: Navigating Super-human Reasoning for Web Agent. *arXiv preprint arXiv:2507.02592* (2025).
- [21] Moxin Li, Yong Zhao, Wenxuan Zhang, Shuaiyi Li, Wenya Xie, See-Kiong Ng, Tat-Seng Chua, and Yang Deng. 2024. Knowledge boundary of large language models: A survey. *arXiv preprint arXiv:2412.12472* (2024).
- [22] Xiaoxi Li, Guanting Dong, Jiajie Jin, Yuyao Zhang, Yujia Zhou, Yutao Zhu, Peitian Zhang, and Zhicheng Dou. 2025. Search-o1: Agentic search-enhanced large reasoning models. *arXiv preprint arXiv:2501.05366* (2025).
- [23] Xiaopeng Li, Pengyue Jia, Derong Xu, Yi Wen, Yingyi Zhang, Wenlin Zhang, Wanyu Wang, Yichao Wang, Zhaocheng Du, Xiangyang Li, et al. 2025. A survey of personalization: From rag to agent. *arXiv preprint arXiv:2504.10147* (2025).
- [24] Xiaopeng Li, Lixin Su, Pengyue Jia, Xiangyu Zhao, Suqi Cheng, Junfeng Wang, and Dawei Yin. 2023. Agent4ranking: Semantic robust ranking via personalized query rewriting using multi-agent llm. *arXiv preprint arXiv:2312.15450* (2023).
- [25] Yuchen Li, Hengyi Cai, Rui Kong, Xinran Chen, Jiamin Chen, Jun Yang, Haojie Zhang, Jiayi Li, Jiayi Wu, Yiqun Chen, et al. 2025. Towards AI Search Paradigm. *arXiv preprint arXiv:2506.17188* (2025).
- [26] Aixun Liao, Bei Feng, Bing Xue, Bingxuan Wang, Bocho Wu, Chengda Lu, Cheng-gang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. 2024. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437* (2024).
- [27] Shuchang Liu, Qingpeng Cai, Bowen Sun, Yuhao Wang, Ji Jiang, Dong Zheng, Peng Jiang, Kun Gai, Xiangyu Zhao, and Yongfeng Zhang. 2023. Exploration and regularization of the latent action space in recommendation. In *Proceedings of the ACM Web Conference 2023*. 833–844.
- [28] Rui Lu, Zhenyu Hou, Zihan Wang, Hanchen Zhang, Xiao Liu, Yujiang Li, Shi Feng, Jie Tang, and Yuxiao Dong. 2025. Deepdive: Advancing deep search agents with knowledge graphs and multi-turn rl. *arXiv preprint arXiv:2509.10446* (2025).
- [29] Shiyu Ni, Keping Bi, Jiafeng Guo, and Xueqi Cheng. 2024. When Do LLMs Need Retrieval Augmentation? Mitigating LLMs' Overconfidence Helps Retrieval Augmentation. In *Findings of the Association for Computational Linguistics ACL 2024*. 11375–11388.
- [30] Judea Pearl. 2009. *Causality*. Cambridge university press.
- [31] Hongjin Qian and Zheng Liu. 2025. Scent of Knowledge: Optimizing Search-Enhanced Reasoning with Information Foraging. *arXiv preprint arXiv:2505.09316* (2025).
- [32] Zile Qiao, Guoxin Chen, Xuanzhong Chen, Donglei Yu, Wenbiao Yin, Xinyu Wang, Zhen Zhang, Baixuan Li, Huifeng Yin, Kuan Li, et al. 2025. WebResearcher: Unleashing unbounded reasoning capability in Long-Horizon Agents. *arXiv preprint arXiv:2509.13309* (2025).
- [33] Peijun Qing, Chongyang Gao, Yefan Zhou, Xingjian Diao, Yaoqing Yang, and Soroush Vosoughi. 2024. Alphasora: Assigning lora experts based on layer training quality. (2024).
- [34] Xin Qiu and Risto Miikkilainen. 2024. Semantic density: Uncertainty quantification for large language models through confidence measurement in semantic space. *Advances in neural information processing systems* 37 (2024), 134507–134533.
- [35] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems* 36 (2023), 53728–53741.
- [36] Evan F Risko and Sam J Gilbert. 2016. Cognitive offloading. *Trends in cognitive sciences* 20, 9 (2016), 676–688.
- [37] Fobo Shi, Peijun Qing, Dong Yang, Nan Wang, Youbo Lei, Haonan Lu, Xiaodong Lin, and Duantengchuan Li. 2024. Prompt space optimizing few-shot reasoning success with large language models. In *Findings of the Association for Computational Linguistics: NAACL 2024*.
- [38] Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Ji-Rong Wen. 2025. R1-searcher: Incentivizing the search capability in llms via reinforcement learning. *arXiv preprint arXiv:2503.05592* (2025).
- [39] Hao Sun, Zile Qiao, Jiayan Guo, Xuanbo Fan, Yingyan Hou, Yong Jiang, Pengjun Xie, Yan Zhang, Fei Huang, and Jingren Zhou. 2025. Zerosearch: Incentivize the search capability of llms without searching. *arXiv preprint arXiv:2505.04588* (2025).
- [40] Qwen Team et al. 2024. Qwen2 technical report. *arXiv preprint arXiv:2407.10671* 2 (2024), 3.
- [41] Hongru Wang, Cheng Qian, Wanjun Zhong, Xiushi Chen, Jiahao Qiu, Shijue Huang, Bowen Jin, Mengdi Wang, Kam-Fai Wong, and Heng Ji. 2025. Acting Less is Reasoning More! Teaching Model to Act Efficiently. *arXiv preprint arXiv:2504.14870* (2025).
- [42] Hongru Wang, Cheng Qian, Wanjun Zhong, Xiushi Chen, Jiahao Qiu, Shijue Huang, Bowen Jin, Mengdi Wang, Kam-Fai Wong, and Heng Ji. 2025. Otc: Optimal tool calls via reinforcement learning. *arXiv e-prints* (2025), arXiv:2504.
- [43] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2023. Text Embeddings by Weakly-Supervised Contrastive Pre-training. In *The Eleventh International Conference on Learning*

- Representations (ICLR)*. https://openreview.net/forum?id=d_h-qG_t-a
- [44] Yejing Wang, Chi Zhang, Xiangyu Zhao, Qidong Liu, Maolin Wang, Xuetao Wei, Zitao Liu, Xing Shi, Xudong Yang, Ling Zhong, et al. 2025. Behavior Modeling Space Reconstruction for E-Commerce Search. In *Proceedings of the ACM on Web Conference 2025*. 1683–1692.
- [45] Ziliang Wang, Xuhui Zheng, Kang An, Cijun Ouyang, Jialu Cai, Yuhang Wang, and Yichao Wu. 2025. StepSearch: Igniting LLMs Search Ability via Step-Wise Proximal Policy Optimization. *arXiv preprint arXiv:2505.15107* (2025).
- [46] Wikimedia Foundation. 2025. English Wikipedia Dump (2025-09-01). <https://dumps.wikimedia.org/enwiki/20250901/>. Accessed: 2025-10-08.
- [47] Yunjia Xi, Jianghao Lin, Yongzhao Xiao, Zheli Zhou, Rong Shan, Te Gao, Jiachen Zhu, Weiwen Liu, Yong Yu, and Weinan Zhang. 2025. A survey of llm-based deep search agents: Paradigm, optimization, evaluation, and challenges. *arXiv preprint arXiv:2508.05668* (2025).
- [48] Derong Xu, Xinhang Li, Ziheng Zhang, Zhenxi Lin, Zhihong Zhu, Zhi Zheng, Xian Wu, Xiangyu Zhao, Tong Xu, and Enhong Chen. 2025. Harnessing large language models for knowledge graph question answering via adaptive multi-aspect retrieval-augmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 25570–25578.
- [49] Derong Xu, Ziheng Zhang, Zhenxi Lin, Xian Wu, Zhihong Zhu, Tong Xu, Xiangyu Zhao, Yefeng Zheng, and Enhong Chen. 2024. Multi-perspective Improvement of Knowledge Graph Completion with Large Language Models. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. 11956–11968.
- [50] Qiancheng Xu, Yongqi Li, Heming Xia, and Wenjie Li. 2024. Enhancing tool retrieval with iterative feedback from large language models. *arXiv preprint arXiv:2406.17465* (2024).
- [51] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388* (2025).
- [52] Dong Yang, Peijun Qing, Yang Li, Haonan Lu, and Xiaodong Lin. [n. d.]. Gamma: Gamma embeddings for logical queries on knowledge graphs. ([n. d.]).
- [53] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 2369–2380.
- [54] Xunjian Yin, Xu Zhang, Jie Ruan, and Xiaojun Wan. 2024. Benchmarking Knowledge Boundary for Large Language Models: A Different Perspective on Model Evaluation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2270–2286.
- [55] Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuan-Jing Huang. 2023. Do Large Language Models Know What They Don't Know?. In *Findings of the Association for Computational Linguistics: ACL 2023*. 8653–8665.
- [56] Chao Zhang, Haoxin Zhang, Shiwei Wu, Di Wu, Tong Xu, Xiangyu Zhao, Yan Gao, Yao Hu, and Enhong Chen. 2025. NoteLLM-2: Multimodal Large Representation Models for Recommendation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*. 2815–2826.
- [57] Dingchu Zhang, Yida Zhao, Jialong Wu, Baixuan Li, Wenbiao Yin, Liwen Zhang, Yong Jiang, Yufeng Li, Kewei Tu, Pengjun Xie, et al. 2025. EvolveSearch: An Iterative Self-Evolving Search Agent. *arXiv preprint arXiv:2505.22501* (2025).
- [58] Wenlin Zhang, Xiangyang Li, Kuicai Dong, Yichao Wang, Pengyue Jia, Xiaopeng Li, Yingyi Zhang, Derong Xu, Zhaocheng Du, Huifeng Guo, et al. 2025. Process vs. Outcome Reward: Which is Better for Agentic RAG Reinforcement Learning. *arXiv preprint arXiv:2505.14069* (2025).
- [59] Wenlin Zhang, Xiangyang Li, Qiyuan Ge, Kuicai Dong, Pengyue Jia, Xiaopeng Li, Zijian Zhang, Maolin Wang, Yichao Wang, Huifeng Guo, et al. 2026. Exploring Recommender System Evaluation: A Multi-Modal User Agent Framework for A/B Testing. *arXiv preprint arXiv:2601.04554* (2026).
- [60] Weinan Zhang, Junwei Liao, Ning Li, Kounianhua Du, and Jianghao Lin. 2024. Agentic information retrieval. *arXiv preprint arXiv:2410.09713* (2024).
- [61] Xiangyu Zhao, Long Xia, Jiliang Tang, and Dawei Yin. 2019. Deep Reinforcement Learning for Search, Recommendation, and Online Advertising: A Survey. *ACM sigweb newsletter* 2019, Spring (2019), 1–15.
- [62] Xiangyu Zhao, Long Xia, Liang Zhang, Zhuoye Ding, Dawei Yin, and Jiliang Tang. 2018. Deep reinforcement learning for page-wise recommendations. In *Proceedings of the 12th ACM conference on recommender systems*. 95–103.
- [63] Xiangyu Zhao, Liang Zhang, Zhuoye Ding, Long Xia, Jiliang Tang, and Dawei Yin. 2018. Recommendations with negative feedback via pairwise deep reinforcement learning. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*. 1040–1048.

A Uncertainty and the Decision Boundary

In this section, we investigate the relationship between the agent's output entropy and its decision boundary errors. We observe the

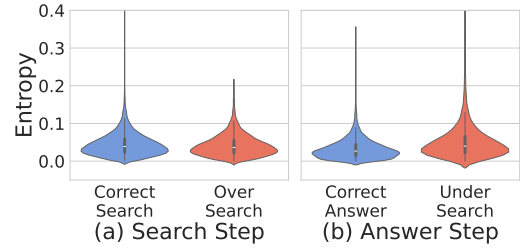


Figure 4: Entropy distributions for correct vs. incorrect decisions at the Search Step (left) and Answer Step (right).

distribution of entropy for correct versus incorrect decisions (over-search and under-search) during both the search and answer steps. The results, shown in Figure 4, reveal the difference in these distributions. As illustrated, for the *Search Step*, the entropy distributions for effective search and over-search are highly similar, indicating little difference. In contrast, for the *Answer Step*, the overall entropy for correct-answer is lower than that for under-search, which suggests the model is more confident when answering correctly.

Furthermore, we examined the potential of using entropy to mitigate decision boundary errors. We treated this as a classification task, using entropy as the feature to distinguish correct from incorrect decisions. The results are presented as ROC curves in Figure 5. As shown, entropy offers almost no contribution to resolving the decision boundary problem for the *Search Step* (AUC = 0.53). However, it provides a discernible, albeit moderate, benefit for the *Answer Step* (AUC = 0.64), indicating its utility in identifying potential under-search errors.

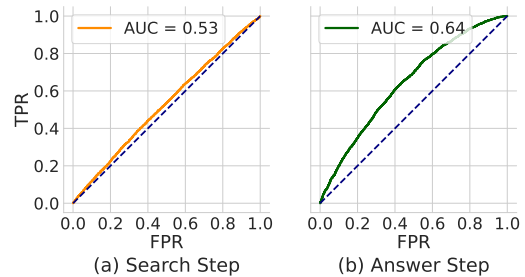


Figure 5: ROC curves illustrating the effectiveness of using entropy to classify correct vs. incorrect decisions for the Search Step (a) and Answer Step (b).

B Prompt Instructions

We list the prompt utilized in our experiments as follows:

- **Intervention Instruction.** Building upon the probing methodology from Section 2.2, we detail the specific instructions used to control the agent's actions. We present two interventions: the *Answer Intervention Instruction* (Figure 7) and the *Search Intervention Instruction* (Figure 6).
- **Knowledge Boundary.** Building upon our discussion of the knowledge boundary in Section 3.5, we now present the specific prompts designed to evaluate an agent's knowledge recall and its meta-knowledge. The *knowledge recall prompt* is shown in Figure 8, and the *knowledge meta prompt* is displayed in Figure 9.

Search Intervention Instruction

You are an advanced reasoning agent tasked with answering a question by searching for information. Follow these steps:

- (1) **Initial Plan:** In your first `<think>` block, decompose the question and formulate a clear, step-by-step plan.
- (2) **Critique & Justify:** After receiving new information, critically evaluate your current knowledge state inside `<think>`. Explicitly answer:
 - What is the specific knowledge gap I need to fill right now?
 - Is my previous reasoning free of errors or hallucinations?
 - Why is another search necessary to solve the problem?
- (3) **Targeted Search:** Based on your critique, formulate a specific, targeted, and non-repetitive search query inside `<search>` to resolve the identified gap.

Do not provide the answer yet. Question: {question}

Figure 6: The system prompt for the *Search Intervention Instruction*. This prompt guides the agent to re-evaluate its knowledge state, identify a specific gap, and formulate a necessary search query instead of answering prematurely.

Answer Intervention Instruction

Answer the given question.

You must conduct reasoning inside `<think>` and `</think>` first every time you get new information.

After reasoning, since you have all the information needed, you must directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations or further searches. For example, `<answer>Beijing</answer>`. Question: {question}

Figure 7: The system prompt for the *Answer Intervention Instruction*. This prompt instructs the agent to bypass the search step and synthesize a final answer using only the knowledge contained within its current state.

Knowledge Recall Prompt

You are acting as a search engine. The following text is a user's raw request, which contains their intended search query inside `<search>` tags. Your task is to directly answer the user's intended query based on your internal knowledge. Provide a concise, direct answer. Do not explain your reasoning or mention that you are an AI.

User's Full Request: {question}

Direct Answer:

Figure 8: The *Knowledge Recall Prompt*. This prompt tests the model's ability to directly retrieve and state factual information from its internal knowledge base, mimicking the behavior of a search engine's direct answer feature.

Knowledge Meta-Cognition Prompt

You are a helpful assistant. You will see a user's full request which contains a search query inside `<search>` tags. Based on the *intended query* within that request, do you believe you have sufficient internal knowledge to answer it accurately without searching? Answer *only* with 'Yes' or 'No'.

User's Full Request: {question}

Answer:

Figure 9: The *Knowledge Meta-Cognition Prompt*. This prompt is designed to probe the model's ability to self-assess the sufficiency of its own knowledge regarding a specific query, forcing a binary 'Yes' or 'No' response.